INCH-POUND
MIL-PRF-20/30D
21 April 2011
SUPERSEDING
MIL-PRF-20/30C
21 June 2001

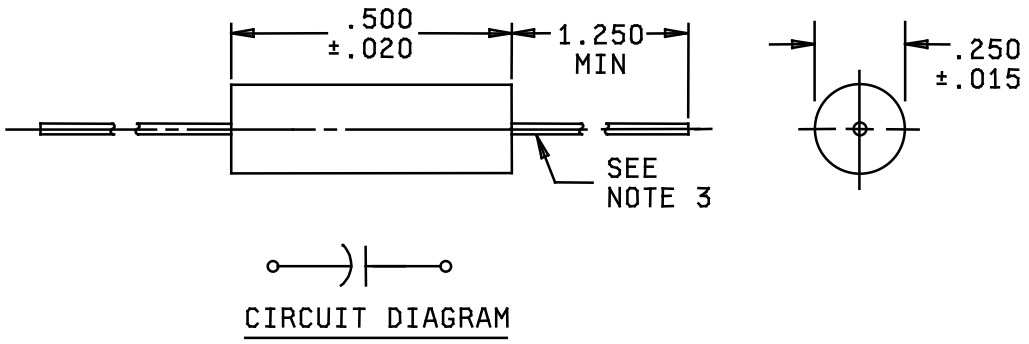
PERFORMANCE SPECIFICATION SHEET

CAPACITORS, FIXED, CERAMIC DIELECTRIC (TEMPERATURE COMPENSATING), ESTABLISHED AND NON-ESTABLISHED RELIABILITY, STYLES CCR78 AND CC78

CC78 is inactive for new design after 21 April 2011. Use CCR78.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the capacitors described herein shall consist of this specification sheet and MIL-PRF-20.



Inches	mm
.015	0.38
.020	0.51
.027	0.69
.250	6.35
.500	12.70
1.250	31.75

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Lead diameter shall be .015 (0.38 mm) to .027 (0.69 mm).
4. Lead length may be a minimum of 1.00 inch (25.4 mm) long for use in tape and reel packaging, when specified in the ordering data.

FIGURE 1. Styles CCR78 and CC78 capacitors.

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REQUIREMENTS:

Dimensions and configuration: See figure 1.

Lead type: Axial.

Case type: Tubular, molded.

DC rated voltage: See table I.

Operating temperature range: -55°C to +125°C.

Characteristic: CG.

TABLE I. Capacitor characteristics.

PIN 1/	Rated voltage (volts, dc)	Nominal capacitance (pF)	Capacitance tolerance	PIN 1/	Rated voltage (volts, dc)	Nominal capacitance (pF)	Capacitance tolerance
CC-78CG821--	200	820	FGJK	CC-78CG562--	100	5,600	FGJK
CC-78CG102--	200	1,000	FGJK	CC-78CG682--	100	6,800	FGJK
CC-78CG122--	200	1,200	FGJK	CC-78CG822--	100	8,200	FGJK
CC-78CG152--	200	1,500	FGJK	CC-78CG103--	100	10,000	FGJK
CC-78CG182--	200	1,800	FGJK	CC-78CG123--	100	12,000	FGJK
CC-78CG222--	200	2,200	FGJK	CC-78CG153--	50	15,000	FGJK
CC-78CG272--	200	2,700	FGJK	CC-78CG183--	50	18,000	FGJK
CC-78CG332--	200	3,300	FGJK	CC-78CG223--	50	22,000	FGJK
CC-78CG392--	100	3,900	FGJK	CC-78CG273--	50	27,000	FGJK
CC-78CG472--	100	4,700	FGJK				

1/ Complete PIN will include the following:
1st dash - Symbol “R” (for style CCR78) or dash will be deleted (for style CC78).
2nd dash - Applicable capacitance tolerance symbol.
3rd dash - Applicable failure rate level symbol (CCR78 only) or dash will be deleted (for style CC78).

Failure rate level (CCR78 only): M (1.0 percent), P (0.1 percent), R (.01 percent) or S (.001 percent).

Thermal shock and voltage conditioning (CCR78 only): In accordance with MIL-PRF-20.

Capacitance: Within tolerance specified (see table I).

Dissipation factor: In accordance with MIL-PRF-20.

Dielectric withstanding voltage: In accordance with MIL-PRF-20.

Body insulation: Test II.

Solderability: In accordance with MIL-PRF-20.

Resistance to soldering heat: In accordance with MIL-PRF-20.

Life: In accordance with MIL-PRF-20, operating condition 2.

Part or Identifying Number (PIN): In accordance with MIL-PRF-20 and table I.

Marking: In accordance with MIL-PRF-20.

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Changes from previous issue: The margins of this specification sheet are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - CR
Navy - EC
Air Force - 85
DLA - CC

Preparing activity:
DLA - CC

(Project 5910-2011-014)

Review activities:
Navy - AS, MC, OS, SH
Air Force - 99

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